

AI-Generated Content Detection System: Project Abstract

Title

Multi-Modal Deep Learning Approach for Detection and Classification of AI-Generated Images and Videos

Abstract

The rapid advancement of generative artificial intelligence models, particularly Latent Diffusion Models (LDMs) such as Stable Diffusion, DALL-E, and Midjourney, has enabled the creation of photorealistic synthetic images that are virtually indistinguishable from authentic photographs by human observers. This technological capability poses significant challenges to data authenticity, digital forensics, cybersecurity, and the prevention of misinformation. The proliferation of AI-generated imagery raises critical concerns regarding privacy violations, fraud, fake news propagation, and the erosion of trust in visual media.

This project proposes a comprehensive AI-generated content detection system that combines multiple analytical approaches to identify synthetic images and videos with high accuracy. The system employs a hybrid architecture integrating Convolutional Neural Networks (CNNs) for spatial feature extraction, frequency domain analysis (Discrete Cosine Transform and Fast Fourier Transform) for detecting digital fingerprints, and specialized artifact detection modules for identifying generation-specific anomalies. The multi-modal approach addresses the limitations of single-method detection systems and provides robustness against evolving generative models.

The methodology involves training deep learning models on a balanced dataset of 60,000 images comprising both authentic photographs and AI-generated synthetic images from various generative models. Transfer learning with pre-trained EfficientNet architectures accelerates training and improves generalization. The system incorporates data augmentation techniques to enhance model robustness and implements explainable AI (XAI) through Gradient-weighted Class Activation Mapping (Grad-CAM) to provide interpretable predictions and visualize regions contributing to classification decisions.

Key innovations include: (1) a hybrid feature fusion approach combining CNN-extracted spatial features with frequency domain signatures, (2) real-time detection capabilities for both static images and video content through frame-by-frame analysis, (3) confidence scoring mechanisms to flag uncertain cases for human review, and (4) extensibility to detect outputs from multiple generative models including Stable Diffusion, Midjourney, and DALL-E.

The system achieves an expected accuracy of 85-92% on test datasets, with precision and recall metrics exceeding 85%, demonstrating its effectiveness in distinguishing AI-generated content from authentic visual media. Comparative analysis with the CIFAKE benchmark study validates the approach, while ablation studies

quantify the contribution of individual detection components. The implementation includes a user-friendly interface deployable as a web application, REST API, or browser extension, facilitating integration into content moderation workflows, digital forensics pipelines, and media verification systems.

This research contributes to the growing field of synthetic media detection by providing an accessible, interpretable, and effective tool for identifying AI-generated visual content. The project addresses critical societal needs for maintaining visual media authenticity, combating misinformation, and protecting against malicious uses of generative AI technology. Future enhancements include expanding detection capabilities to audio deepfakes, implementing adversarial robustness testing, and developing continuous learning mechanisms to adapt to emerging generative models.

Keywords: Artificial Intelligence, Deep Learning, Image Classification, Generative Models, Deepfake Detection, Computer Vision, Convolutional Neural Networks, Explainable AI, Digital Forensics, Cybersecurity

Project Highlights

Problem Statement

Detection of AI-generated images and videos to prevent misinformation, fraud, and maintain digital media authenticity in an era where synthetic content is indistinguishable from reality.

Objectives

1. Develop a multi-modal detection system combining CNN and frequency analysis
2. Achieve $\geq 85\%$ accuracy on diverse AI-generated content
3. Provide explainable predictions through XAI visualization
4. Support real-time detection for images and videos
5. Create deployable tools (API, web interface, browser extension)

Methodology

- **Data Collection:** 60K balanced dataset (real + AI-generated images)
- **Architecture:** Hybrid CNN + Frequency Analysis + Artifact Detection
- **Training:** Transfer learning with EfficientNet, data augmentation
- **Evaluation:** Accuracy, Precision, Recall, F1-Score, AUC-ROC
- **Interpretability:** Grad-CAM for visual explanations

Expected Outcomes

- **Detection Accuracy:** 85-92%

- **Precision/Recall:** >85%
- **Processing Speed:** <1 second per image
- **Video Analysis:** 30 frames per second capability
- **Deployable System:** Web app, API, browser extension

Applications

- Social media content moderation
- News and journalism verification
- Digital forensics and legal evidence
- Academic integrity checking
- Cybersecurity threat detection

Technical Contributions

1. Novel multi-modal fusion architecture
2. Comprehensive benchmark on multiple AI generators
3. Explainable AI implementation for transparency
4. Open-source toolkit for research community
5. Practical deployment strategies

Societal Impact

- Combat misinformation and fake news
 - Protect privacy and prevent fraud
 - Maintain trust in visual media
 - Support digital forensics investigations
 - Promote ethical AI development
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Research Significance

This project addresses a critical gap in current cybersecurity and digital forensics capabilities. While generative AI models continue to improve, detection mechanisms must evolve correspondingly. The multi-modal approach provides resilience against single-point failures and offers better generalization across different generative architectures. The inclusion of explainable AI components ensures transparency and trust in automated detection systems, which is essential for legal and forensic applications.

The research contributes both theoretical insights into AI-generated content characteristics and practical tools for real-world deployment. By making the system accessible through multiple interfaces (API, web app, extension), the project enables widespread adoption and integration into existing workflows. The open-source release of code and findings promotes collaborative advancement in this critical area of AI safety and security.

As synthetic media generation becomes increasingly sophisticated, this project provides a foundation for ongoing research and development in detection technologies, helping society navigate the challenges posed by artificial intelligence while maximizing its benefits.

Project Duration

Estimated Timeline: 10-12 weeks

Team Requirements

- 1-3 researchers/developers
- Access to GPU computing resources
- Domain expertise in deep learning and computer vision

Deliverables

1. Trained detection models (multiple architectures)
 2. Complete source code repository
 3. Technical documentation and API reference
 4. Research paper/thesis documenting findings
 5. Deployment-ready applications (web, API, extension)
 6. Dataset and benchmarks (if permissible)
 7. Demo video and presentation materials
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Contact Information: [Your Name/Institution] [Project Repository: GitHub URL] [Documentation: Project Website]

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